Core Python Test

(1) Find Valid identifiers
   a. _my_var
   b. 2nd_value
   c. total$
   d. while
   e. attempt_2
   f. __abc

(2) How many reserved keywords in python? Write python code to print all reserved keywords?

(3) What is Data type and How many data types are in python?

(4) What is Type Casting in python?

(5) How many fundamentals data types are in python?

(6) What is difference between Mutable & Immutable?

(7) Which of the following statements accurately describes Python's fundamental data types?
   A) All fundamental data types are mutable.
   B) All fundamental data types are immutable.

(8) What is the difference between Normal Division (/) and Floor Division (//)?
   Provide output: -
   a. print (10/3)
   b. Print (10//3)
   c. Print(10/4)
   d. Print(10.0//4)
   e. Print(10/4)

(9) What is string concatenation? Give example.

(10) What is the difference between == and = operator? Give example

(11) What will be the output?

```
X = (10==10)
Print(x)
```

(12) What will be the result of the following expressions?

a. "Python" and "Java"
b. "" and "Python"
c. "Java" and ""
d. "" or "BTech"
e. "BTech" or ""
f. "Python" or "Java"

(13) What will be the result of the following expressions? Write the name of operator as well?

a. 10 << 3
b. 40 >> 2

(14) Explain with example, what is identity & membership operator?

(15) Write a Program to find Biggest of given 3 Numbers from the Command Prompt?

(16) Write a python program to compute the sum of the first 10 natural numbers.

a. Using for loop
b. Using while loop

(17) What will be the output?

```
cart=[10,20,600,60,70]
 for item in cart:
   if item>500:
     print("To place this order insurance must be required")
     break
   print(item)
```

(18) What will be the output?

```
for i in range(10):
if i%2==0:
   continue
print(i)
```

(19) What will be the output?

```
text = "Python Programming"
```

a. print(text[0:6])
b. print(text[:6])
c. print(text[7:])
d. print(text[::2])
e. print(text[-3:])
f. print(text[::-1])

(20) Write a python program to take string as “Python” from user and reverse it.

(21) S = “python is hard language”
Using string implementation print
S = “python is easy language”

(22) S = “python is hard language”
Write a python program to reverse.

S= “esrever drah si nohtyp”

(23) Write a program for pattern?

```
A
A B
A B C
A B C D
A B C D E
A B C D E F
A B C D E F G
A B C D E F G H
A B C D E F G H I
A B C D E F G H I J
```

hint-> chr(64+1) -> A

(24) Write a program for pattern?

```
1
2 3
4  5 6
7   8  9  10
```

(25) Write a Program to display Unique Vowels present in the given Word using list?

Word -> “pyaathnidfrenju”

(26) Write a Program to take a Tuple of Numbers from the Keyboard and Print its Sum and Average?

(27) Write difference between list and tuple?

(28) Write a python Program to Enter Name and Percentage Marks of student in a Dictionary and Display Information on the Screen?

(29) Write a Program to find Number of Occurrences of each Letter present in the given String?

Output should be like below:-

```
Enter any word: mississippi
    m occurred  1  times
    i occurred  4  times
    s occurred  4  times
    p occurred  2  times
```

(30) Write a function for factorial of a number.

a. Factorial of 0 is 1
b. Factorial of negative number should print any statement
c. Greater than 1 value should return value.

(31) What is variable length parameter? Explain with example.

(32) Using list comprehension, print square of number.

(33) Using dictionary comprehension, print table of 15.

(34) What is anonymous function? Write python program to find maximum of two numbers using anonymous function?

(35) What is the use of filter () and map () function?

(36) Write a python program to swap two numbers? With and without third variable.

(37) Write a python program to generate random numbers between 10 to 100.

(38) What is modules and packages in python?

(39) Write a python program to check any year is leap year or not?

(40) Given s = “abc”

Write a python program for s = aababc

(41) Which of the following are valid Tuples?

(a) t = ()
(b) t = 10, 20, 30, 40
(c) t = 10
(d) t = 10,
(e) t = (10)
(f) t = (10,)
(g) t = (10, 20, 30, 40)

(42) Given string s = “PYTHON”

How to convert string as list of strings?

li = [‘P’, ‘Y’, ‘T’, ‘H’, ‘O’, ‘N’]

(43) Write a python code for check whether the number is prime number?
Prime number is only divisible by 1 and same number.

(44) Li = [12, 45, 78, 23, “python”, “java”]

a. Add “Typescript” in list
b. Add “Typescript” at 2nd index
c. Append another list, s = [100, 101, 102] inside li list
d. Print new li list
e. Remove last element inside li list
f. Reverse the list
g. Sort the list
h. Print the new list
i. Make list as string and print it

(45) t = (12, 45, 67, 56)

a. add 234 in t
b. add 345 in t
c. remove 45 from t
d. sort elements inside t

(46) s = “ I am Python “

a. remove left and right space in string
b. remove only left space

(47) li = [1, 2, 4, 67, 34, 56, 66, 123, 34, 34, 34, 56, 56, 56]

a. Convert list data into set, assign a new variable of set and print.

(48) What is the difference between `*args` & `**kwargs?`

(49) **What is the difference between Modules and Packages, Give some example?**

(50)

```
d = {'apples': 20, 'bananas': 15, 'oranges': 10, 'grapes': 5}

a. Add one new fruit to d. ex: 'pomegranate':67
b. Print all keys separately
c. Print all values separately
d. Remove apples from dict and print remaining fruits inside d.
e. Fetch oranges value
```

(51) k = ["A", "B", "C", "D","E"]
v = [65, 66, 67, 68, 69]

Both K and V are list.

Using both, I want to create a dictionary.